Team Name: NsquareHer

Project Name: LifeLight

WePower - Empowering women in clean energy

SDG Goal Focus: 7

Team Members

Nishita Saxena

Nidhi Pareek

Problem Statement:

- Around 1 billion people depend on healthcare facilities with unreliable electricity, affecting critical medical services. (WHO – Energizing Health Report, 2023)
- 260,000 women die annually from pregnancy and childbirth complications, with unreliable power worsening maternal care in rural clinics. (WHO Maternal Mortality Fact Sheet)
- 21,000+ healthcare facilities now use solar energy, but solar power drops 23–67% in cloudy conditions and produces 0 power at night.
- 1 in 3 solar health systems fail due to poor monitoring and energy management.
- During shortages, healthcare workers must manually decide which medical devices stay powered.
- 90% of nurses and midwives are women, yet they receive no training in managing clinic energy systems.
- World Bank aims to solarise 100,000 healthcare facilities by 2030, increasing the need for intelligent energy management systems that both optimize clean energy use and empower frontline women healthcare workers.

Introduction:

- LifeLight is a low-cost AI-powered solar energy management system for off-grid maternity clinics.
- Runs on embedded hardware (<$80) for scalable deployment.
- Monitors solar generation and clinic energy usage in real time.
- Automatically protects critical medical equipment during power shortages.
- Generates simple weekly AI reports explaining energy patterns and efficiency improvements.
- Designed to build energy literacy for frontline women healthcare workers, enabling them to confidently manage their clinic’s energy systems.

Research Survey:

Existing Energy Management Systems

- Commercial EMS platforms (SolarEdge, Victron, Schneider Electric) are built for grid-connected residential or industrial systems, requiring technical setup and trained operators — not suitable for off-grid rural clinics.

Non-Intrusive Load Monitoring (NILM)

- NILM has been widely studied since Hart (1992) for residential and commercial energy monitoring, but no implementation targets rural healthcare facilities.

Solar Forecasting with LSTM

- Research shows LSTM models outperform traditional forecasting methods for short-term solar prediction and can run efficiently on edge devices like Raspberry Pi.

Health Worker Knowledge Gap

- Studies from WHO, UNDP, and solar health programs identify lack of energy literacy among healthcare workers as a key barrier, recommending monitoring and training tools.

Operational Failures in Solar Clinics

- Studies in Ethiopia, Ghana, and Uganda report frequent solar system failures due to absence of monitoring, maintenance, and load prioritization.

Proposed Solution:

- Dual-Hardware Intelligent System.
  - ESP32 → Real-time energy monitoring and control.
  - Raspberry Pi Zero 2W → AI analysis and forecasting

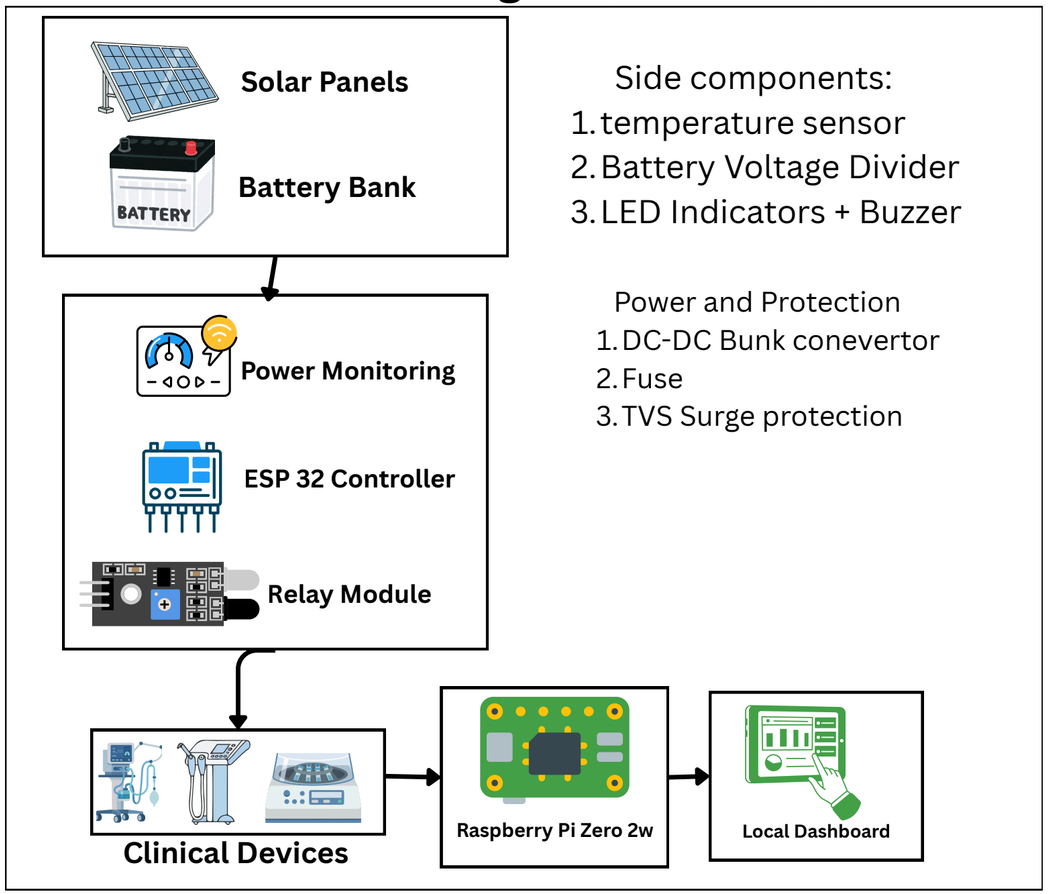
- Smart Energy Management:
  - Non-Intrusive Load Monitoring identifies active devices using a single current sensor.
    - AI model forecasts solar generation 4 hours ahead to anticipate shortages.
    - Critical medical loads remain protected via fail-safe circuits.
    - Non-critical devices are automatically managed to maintain energy stability.

- Human-Centered Control:
  - If safe automation is not possible, the system alerts the healthcare worker with a voice prompt and real-time energy dashboard, enabling informed decisions.

- Energy Education Layer:
  - Weekly AI-generated reports explain clinic energy usage in simple language, helping women healthcare workers develop technical understanding and energy leadership.

Prototype (if applicable):

Architecture Diagram(hardware):



Side components:

1. temperature sensor
2. Battery Voltage Divider
3. LED Indicators + Buzzer

Power and Protection

1. DC-DC Bunk conevertor
2. Fuse
3. TVS Surge protection

Local dashboard live demo(software):

https://lifelight-one.vercel.app/

Advantages

- Fail-Safe Design – Normally-open relays keep critical devices powered even if the controller fails.
- Proactive Energy Management – LSTM forecasting predicts energy shortages hours in advance.
- Low-Cost Monitoring – Single-sensor NILM monitors all devices, reducing cost and complexity.
- Dual-Layer Reliability – ESP32 maintains real-time control even if the Raspberry Pi fails.
- Energy Literacy – Weekly reports help healthcare workers understand and manage energy use.
- Offline Operation – Local WiFi dashboard runs without internet or cloud services.

Disadvantages:

- NILM Accuracy Limits – Simultaneous device activation can reduce load identification accuracy.
- SOC Estimation Errors – Battery aging affects Coulomb-counting accuracy, requiring conservative energy decisions.
- Internet Needed for Reports – LLM-based reports require occasional internet; offline templates provide basic output.
- Single Sensor Dependency – Failure of the main power sensor removes device-level monitoring.
- Initial Calibration – Each clinic requires one-time device signature calibration.

Conclusion:

Solar electrification without intelligent energy management is an incomplete solution. When generation drops and medical devices compete for limited battery reserves, the consequences fall on the healthcare worker and on her patients.

LifeLight AI closes this gap. It prioritizes critical equipment autonomously, anticipates shortages before they occur, and operates without internet, technical expertise, or expensive infrastructure at under $80.

Beyond energy management, it builds the energy literacy of the healthcare worker herself so that over time, she understands her clinic's power system well enough to manage it confidently on her own.

As 100,000 healthcare facilities transition to solar by 2030, intelligent energy management is not optional.

References:

Healthcare Energy Access & Infrastructure

- World Health Organization - Maternal Mortality Fact Sheet, 2023
- World Health Organization / World Bank / IRENA - Energizing Health Report, 2023
- World Bank / IRENA - COP27 Solar Electrification Pledge, 2022
- UNDP - Solar for Health: Implementation Field Reports, 2021

Solar System Failures

- PMC - Solar PV Performance in Variable Conditions
- WHO / Energizing Health Partners - Solar Health System Failures
- WHO / PMC — Ghana-Uganda Solar Research

Technical Foundations

- Hart, G. W. (1992) - Nonintrusive Appliance Load Monitoring
- LSTM-Based Solar Forecasting - JISEM Journal
- Reliability and Maintainability Analysis of Solar Photovoltaic Systems in Rural Health Facilities
- Solar Powering Public Health Centers: A Systems Thinking Lens

Thank You